

Introduction to SPTI+ and OSINT

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Agenda

- whoami
- Game rules
- Introductions
- Small intro to cybersecurity
- Curriculum
- OSINT

whoami



- Ingeniero de Sistemas
- Máster en Informática con Énfasis en Ciberseguridad



Game rules

- All labs have the same weight in the final grade
 - The deliverable will always be the same: a report
 - Almost everything will be in English (as you can probably tell by now)
 - Work will always be in pairs, and the pairs will always be the same.

Game rules

- Moodle
 - Homework
 - Class materials
- Teams
 - Announcements for next class.
 - Questions to me!
- Attendance and participation

Curriculum

1. Introduction to cybersecurity and OSINT
2. Linux
3. Reporting
4. Threat Modeling
5. Reverse Engineering
6. Cryptography
7. Malware Analysis
8. Vulnerability Management
9. Ethical Hacking
10. Buffer Overflow
11. Network Analysis
12. Security Architecture
13. Digital Forensics
14. Application Security
15. Scripting
16. DevSecOps

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Or through Teams!



Cybersecurity

Cybersecurity is the set of practices, technologies, and processes designed to protect computer systems, networks, programs, and data from attacks, unauthorized access, or damage.

The cybersecurity triad



- **Confidentiality:** There should be no unauthorized access to information.
- **Integrity:** There should be no unauthorized modification of the information.
- **Availability:** Authorized users should be able to access information whenever they need it.



RED TEAM

ATTACKERS

Works to break into the system.

Skills include:

- Penetration Testing
- Vulnerability Scanning
- Social Engineering
- Threat Intelligence
- Custom Toolset Development

PURPLE TEAM

MEMBERS FROM BOTH TEAMS

Gets blue and red teams to work together to improve an organisation's security posture.

Skills include:

- Collaboration
- Information-Sharing
- Reporting and Analysis



BLUE TEAM

DEFENDERS

Works to keep the systems safe.

Skills include:

- Network Monitoring
- Data and Log Analysis
- Risk Assessments
- Threat Detection

Hats



OSINT

What is OSINT?

OSINT (Open Source Intelligence) refers to the process of collecting and analyzing publicly available information in order to obtain useful knowledge.

What is OSINT?

- It is based on accessible sources without the need for invasive techniques.
- It is legal if carried out without violating privacy policies or terms of service.
- It is used in both defensive contexts (audits, threat hunting) and offensive contexts (red teaming, cybercrime).

What is OSINT?

- Search engines (Google, Bing, Yandex)
- Social networks (LinkedIn, Facebook, Twitter, Instagram)
- Public records (Whois, government databases)
- Metadata in documents or images
- Forums, blogs, paste sites (Pastebin, Reddit)
- Exposed domains and web services

OSINT's Importance in Cybersecurity

OSINT is the first phase of the attack chain (Kill Chain), specifically the reconnaissance stage.

Passive vs. Active Recon

- **Passive:** information is collected without directly interacting with the target (stealthier).
- **Active:** interaction with the target's infrastructure or services (riskier).

Advantages of Passive Recon

- Low risk of detection
- Large volumes of information
- Allows you to build a complete profile of a person or organization.

OSINT in real life

- Attackers use it to choose vulnerable targets or for social engineering.
- Security teams use it to identify unintentional public exposures (shadow IT, data leaks, bad staff habits).

OSINT tools in Kali Linux

Tool	Description
theHarvester	Harvests
Maltego CE	
whois	
dig	
dnsrecon	
ExifTool	
Google Dorks	

OSINT techniques for people

Objective: to obtain data that allows a person to be identified, located, or profiled.

Types of information sought:

- Social media profiles
- Emails
- Resumes or online publications
- Photographs with metadata
- Participation in forums or paste sites

Common techniques:

- Advanced search with operators in Google (Google Dorks)
- Queries on sites such as Hunter.io or HaveIBeenPwned
- Reverse image search (Google, Yahoo)

OSINT techniques for infrastructure

Objective: to understand how an organization's digital infrastructure is structured and how exposed it is to the public.

Types of information sought:

- WHOIS domain information
- DNS and subdomains
- IP address and geolocation
- Public emails and technical employees
- Exposed services, administrative roles

Common techniques:

- WHOIS and DNS queries (dig, dnsrecon)
- Email enumeration with theHarvester
- Visualization in Maltego
- Use of search engines and tools such as Shodan or Censys

Ethical and legal considerations

Although OSINT is based on public sources, **not everything that is accessible is legal to use**. Some key considerations:

- **Consent and context:** posting something on social media does not imply consent for its automated collection.
- **Legitimate purpose:** in audits, there must be authorization from the client or the simulated environment.
- **Terms of service:** many platforms explicitly prohibit scraping.
- **Avoid harm:** the privacy and security of third parties must be respected.

Ethical and legal considerations

In professional cybersecurity, ethics and legality are as important as technical knowledge.

Gracias